THREE SCOPED PRODUCT AND CONTRACTION RESULTS FIXED QUADRATIC, SMOOTH CONVEX, AND AFFINE SCALAR REGIMES

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ABSTRACT

This theory-only note records three separate, scoped identities. A fixed quadratic has an exact matrix-product solution whose first-order log contraction depends on cumulative learning-rate exposure, with a finite-step squared-step remainder. Under explicit pointwise ambient Hessian bounds on every minimizer-to-iterate segment, gradient descent on a twice continuously differentiable objective contracts full parameter error; a faster directional rate also requires an invariant-subspace condition. An affine scalar conditional-mean recursion has an exact product law and an exponential upper bound. The statements do not compare two paths and are not laws for logistic regression, shuffled SGD, CNNs, nonconvex optimization, or real label noise. The package makes no empirical, deployment, or novelty claim.

1 INTRODUCTION

This paper is a scoped formal exposition of three product/contraction calculations. It does not define an alternative trajectory, a reference trajectory, or a quantity measuring their difference. Accordingly, it does not establish a comparator-path result, data attribution, or a clean-path return identity. Those topics motivate possible future research but are not conclusions of the results below.

The three results have deliberately separate domains. T1 is a fixed-quadratic product identity. T2 is a smooth-convex full-norm contraction theorem under pointwise ambient Hessian bounds. T3 is an affine scalar conditional-mean identity. They share a product form but no theorem connects their state spaces or consequences. Neither T1 nor T3 establishes a law for nonlinear SGD, and T2 does not transfer automatically to arbitrary neural networks.

Result	State and assumptions	Conclusion	Not supplied
T1	Fixed SPD quadratic; $0 \le \eta_t L < 1$	Exact terminal matrix product and a log-product remainder	A changed objective, a comparator path, or nonlinear transfer
T2	Gradient descent; pointwise $\mu I \preceq \nabla^2 L_A \preceq LI$ on each minimizer–iterate segment	Full-norm contraction to $w_A^\star$; directional rate only with invariance	A comparison to another trajectory
T3	Affine scalar recursion with conditional-zero-mean innovation	Exact mean product and exponential upper bound	A convex-optimization or general-SGD statement

Scope. The frozen public baseline contained historical numerical material without co-located public code, output, configuration, seed, and preregistration artifacts. That material is not part of this submission source. A separate, uncompiled archive points to the immutable checkpoint containing it. No unexecuted empirical policy is part of the active paper.

2 NOTATION

Write $\eta_t \geq 0$ for a deterministic step size and $E_{s:T} = \sum_{t=s}^{T-1} \eta_t$ for tail exposure. The symbol A in T2 only labels the displayed objective L_A; it does not encode a data block or a reference path. "Contraction" below always means the particular norm or expectation displayed in its own theorem, not accuracy, a representation, or a deployment outcome.

3 THREE BOUNDED RESULTS

3.1 T1: FIXED-QUADRATIC EXPOSURE PRODUCT

Lemma 1 (T1). *Let*

$$Q(w) = \tfrac{1}{2}(w - w^\star)^\top H(w - w^\star), \qquad 0 \prec \mu I \preceq H \preceq LI,$$

and run $w_{t+1} = w_t - \eta_t \nabla Q(w_t)$ with $0 \leq \eta_t L < 1$. Then

$$w_T - w^\star = \prod_{t=0}^{T-1} (I - \eta_t H)(w_0 - w^\star). \tag{1}$$

For an eigenvalue λ_j of H, set $P_j = \prod_t (1 - \eta_t \lambda_j)$, $E = \sum_t \eta_t$, and $q_j = \max_t \eta_t \lambda_j < 1$. Then

$$0 \leq -\log P_j - \lambda_j E \leq \frac{\lambda_j^2 \sum_t \eta_t^2}{2(1 - q_j)}. \tag{2}$$

Proof. Since $\nabla Q(w) = H(w - w^\star)$, one step gives $w_{t+1} - w^\star = (I - \eta_t H)(w_t - w^\star)$, and iteration proves equation 1. Diagonalize the symmetric positive-definite H. On an eigenvector with eigenvalue λ_j, the multiplier is P_j. For $x \in [0, q_j]$,

$$0 \leq -\log(1 - x) - x = \sum_{k \geq 2} \frac{x^k}{k} \leq \frac{x^2}{2(1 - q_j)}.$$

Apply this inequality to $x = \eta_t \lambda_j$ and sum over t. This proves equation 2. $\square$

T1 says that equal exposure yields the same first-order log contraction in this fixed quadratic. It does not say that arbitrary finite schedules with equal exposure have identical terminal iterates: the squared-step term quantifies the permitted discrepancy. Replacing a step η by K steps of size η/K preserves exposure, whereas repeating K original steps changes exposure and is not a finite-budget invariance.

3.2 T2: AMBIENT-HESSIAN GRADIENT-DESCENT CONTRACTION

Theorem 1 (T2). *Let L_A be twice continuously differentiable with unique minimizer $w_A^\star$ and $\nabla L_A(w_A^\star) = 0$. Suppose that for every $t = s, \ldots, T-1$ and every $r \in [0, 1]$,*

$$\mu I \preceq \nabla^2 L_A(w_A^\star + r(w_t - w_A^\star)) \preceq LI, \qquad 0 < \mu \leq L. \tag{3}$$

If gradient descent on L_A is run from the arbitrary starting iterate w_s with $0 \leq \eta_t \leq 1/L$, then

$$\|w_T - w_A^\star\|_2 \leq \prod_{t=s}^{T-1} (1 - \eta_t \mu) \|w_s - w_A^\star\|_2 \leq e^{-\mu E_{s:T}} \|w_s - w_A^\star\|_2. \tag{4}$$

If, in addition, $e \neq 0$, $\lambda_e > 0$, and $\mathrm{span}(e)$ is invariant for every segment average Hessian in the proof and its eigenvalue on e is at least λ_e, then

$$|e^\top (w_T - w_A^\star)| \leq e^{-\lambda_e E_{s:T}} |e^\top (w_s - w_A^\star)|. \tag{5}$$

Proof. For each $t \ge s$, the fundamental theorem of calculus gives

$$\nabla L_A(w_t) - \nabla L_A(w_A^\star) = \bar{H}_t(w_t - w_A^\star), \qquad \bar{H}_t = \int_0^1 \nabla^2 L_A(w_A^\star + r(w_t - w_A^\star))\, dr.$$

Integrating the pointwise ambient bounds in equation 3 implies $\mu I \preceq \bar{H}_t \preceq LI$. Therefore

$$w_{t+1} - w_A^\star = (I - \eta_t \bar{H}_t)(w_t - w_A^\star).$$

All eigenvalues of $I - \eta_t \bar{H}_t$ lie in $[0, 1 - \eta_t\mu]$ because $\eta_t \le 1/L$. Hence its spectral norm is at most $1 - \eta_t\mu$. Multiply these bounds from s through $T-1$, then use $1 - x \le e^{-x}$ to obtain equation 4.

For the directional statement, the added invariance premise makes e an eigenvector of every $\bar{H}_t$. Its scalar multiplier is at most $1-\eta_t\lambda_e$; the same multiplication and exponential bound prove equation 5. A Rayleigh-quotient lower bound alone does not imply this shared invariant-subspace premise. □

Thus T2 is a full-parameter contraction result under the explicit pointwise ambient Hessian condition equation 3. It does not compare w_t to a second trajectory or identify why w_s has its value. Its directional refinement is conditional and must not be inferred from observations in another model class.

3.3 T3: AFFINE SCALAR CONDITIONAL-MEAN PRODUCT

Theorem 2 (T3). *Let (u_t) be an integrable scalar process adapted to $(\mathcal{F}_t)$. Fix $\lambda > 0$ and deterministic step sizes satisfying $0 < \eta_t\lambda < 1$. Suppose*

$$u_{t+1} = (1 - \eta_t\lambda)u_t + \eta_t\xi_t, \qquad \mathbb{E}[\xi_t \mid \mathcal{F}_t] = 0. \tag{6}$$

Then for any $s < T$,

$$\mathbb{E}[u_T] = \mathbb{E}[u_s] \prod_{t=s}^{T-1} (1 - \eta_t\lambda), \qquad |\mathbb{E}[u_T]| \le |\mathbb{E}[u_s]| e^{-\lambda E_{s:T}}. \tag{7}$$

Proof. Taking conditional expectations in equation 6 and then total expectations yields

$$\mathbb{E}[u_{t+1}] = (1 - \eta_t\lambda)\mathbb{E}[u_t].$$

Repeated substitution gives the product identity in equation 7. Every factor is positive and $1 - x \le e^{-x}$ for $x \ge 0$, so multiplying the inequalities gives the exponential upper bound. □

T3 is exact for the displayed scalar recursion. It does not assert that logistic-SGD, random reshuffling, or a nonlinear accuracy functional obeys equation 6.

4 FUTURE WORK

No empirical policy or experiment is defined or evaluated in this paper. Possible future work could formulate an explicit comparator-path theorem or preregister and execute a separately defined empirical study. Such work would require its own inputs, comparator, and evidence chain; it would test practical relevance, not serve as evidence for T1–T3.

5 RELATED WORK

The locked sources on training-data/model relationships—influence functions (Koh & Liang, 2017), datamodels (Ilyas et al., 2022), TRAK (Park et al., 2023), and SISA (Bourtoule et al., 2021)—are contextual rather than antecedents for the three results. Work on SGD noise and implicit regularization (Mandt et al., 2017; Smith & Le, 2018; Smith et al., 2021) supports stating T3's affine conditional-mean premise instead of treating it as a general SGD identity. The currently locked sources do not provide a verified theorem-level antecedent comparison for all of T1–T3. We therefore make no claim that the individual identities are novel, nor that their juxtaposition is a nontrivial unifying result. The closest-work matrix records this boundary and the remaining positioning gap.

6 Limitations

The scope is mathematical rather than empirical. T1 is fixed quadratic. T2 requires the pointwise ambient spectral Hessian bounds equation 3 on every relevant minimizer–iterate segment, a stable step-size bound, and a separate invariant-subspace condition for its directional refinement. T3 requires the affine scalar recursion and conditional-zero-mean innovation. None of these assumptions automatically holds for logistic training, shuffled SGD, neural networks, noisy labels, or data-selection systems. The public package also lacks a same-version experiment evidence chain; remote candidate results are not attributed to this manuscript.

The contribution ceiling is also substantive: without a theorem that connects the three regimes, a verified theorem-level distinction from direct antecedents, or new empirical evidence, this package remains a scoped exposition rather than an established research advance.

7 Conclusion

T1–T3 establish exact or bounded product relations only within their separate stated regimes. They do not establish a comparator-path identity, a training-path effect, or a deployment recommendation. A future paper may add a common formal model or empirical evidence; this one does not.

Reproducibility Statement

This self-contained source package includes its consumed TeX dependencies and bibliography. The controlled build record pins the compiler and BibTeX versions, hashes every consumed source input (including this statement), and binds a clean rebuild to the archived PDF. Formal assumptions, statements, and proofs are active in this source. No unexecuted protocol or remote-only result is reported as an observation.

Ethics Statement

This revision reports no newly run human-subject or dataset study. Any future execution must verify dataset licenses, versions, and use conditions, and the policy must not access clean, protected, test, or oracle labels at decision time.

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- Drafting and revising prose, equations, figures, tables, and this disclosure; generating alternative paper stories and revision plans.
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A PROOF-SCOPE CHECKLIST

T1 uses a fixed symmetric positive-definite Hessian and its eigendecomposition. T2 uses the pointwise ambient bounds equation 3, whose integration bounds every segment-average Hessian between μI and LI; its directional line is conditional on simultaneous invariance. T3 takes conditional expectations of the stated scalar recursion. These are the only proof dependencies used by the active manuscript. Historical figures, tables, scripts, and numerical claims are not imported, hidden, or compiled here.